

# An Arbitrary-Family Diagonal Principle for Nets

Candidate full solution for arXiv:2304.04189

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**Status.** Candidate full solution, likely valid pending expert review. The proof is in ZFC and explicitly uses choice; it is not claimed to be constructive.

## 1 Source question

Azouzi proves a diagonal principle for a countable sequence of properties of nets [1]. At the start of Section 4, PDF page 7, he asks whether there is “a more general diagonal process where the set of properties is not assumed countable,” observes that a constructive proof should not be expected, and suggests Zorn’s lemma. The complete source passage is shown below.

## 4 Examples of application

We present in this section a few results as application of our main Theorem. The two first applications are just new short proofs of known results in topology and functional analysis. The two next applications are new results and concern compact operators, answering open questions raised recently in [3] and [1], where authors treated the sequel case following classical diagonal processes for sequences and left open the general case because of the lack of a similar result, such result is provided by Theorem 7. It is natural to ask if there is a more general diagonal process where the set of properties is not assumed countable. In that case a constructive proof can not be expected, and a use of Zorn’s lemma seems to be evident. This kind of result could be applied in more general situations and allows to get a very short proofs for Tychonoff Theorem, Alaoglu theorem and several other results in topology and functional analysis. We plan to explore this idea in future work.

The natural precise formulation is obtained from Theorem 7 of the source by replacing  $(\mathcal{P}_k)_{k \in \mathbb{N}}$  with a family indexed by an arbitrary set.

## 2 Definitions

We use the subnet convention of the source:  $y: B \rightarrow E$  is a subnet of  $x: A \rightarrow E$  if  $y = x \circ \phi$  for a *final* map  $\phi: B \rightarrow A$ , meaning that for every  $\alpha_0 \in A$  there is  $\beta_0 \in B$  such that  $\phi(\beta) \geq \alpha_0$  whenever  $\beta \geq \beta_0$ . It is an *eventual subnet* if some tail is a subnet.

For a net  $x$  in  $E$ , its eventuality filter is

$$\mathfrak{F}_x = \{S \subseteq E : x \text{ is eventually in } S\}.$$

The net is *universal* if, for every  $S \subseteq E$ , it is eventually in  $S$  or eventually in  $E \setminus S$ . Equivalently,  $\mathfrak{F}_x$  is an ultrafilter.

### 3 Proof intuition

Countable diagonalization repeatedly refines a net and then follows one diagonal through the resulting chain. For an arbitrary family there is no last stage and no elementary transfinite limit subnet. The replacement is to make the tail behavior maximal first.

Pass to a universal subnet  $u$ . Every further subnet of  $u$  has exactly the same eventuality ultrafilter. For each desired property choose a satisfying subnet  $v^i$  of  $u$ . Because all their tail ranges belong to one ultrafilter, every finite collection of tail requirements has a common value. Indexing over all finite collections of requirements produces one net which is a subnet of  $u$  and, on a tail, a subnet of every  $v^i$ . Heredity under eventual subnets then transfers all properties at once.

### 4 Two lemmas

**Lemma 1** (Universal subnet). *Every net has a universal subnet, assuming the ultrafilter lemma.*

*Proof.* Let  $x: A \rightarrow E$  and put  $T_\alpha = \{x_\gamma : \gamma \geq \alpha\}$ . The sets  $T_\alpha$  form a filter base. Extend the generated filter to an ultrafilter  $\mathcal{U}$  on  $E$ .

Let

$$D = A \times \mathcal{U}, \quad (\alpha, V) \preceq (\alpha', V') \iff \alpha \leq \alpha' \text{ and } V' \subseteq V.$$

This is directed: use an upper bound in  $A$  and the intersection of the two ultrafilter sets. Since  $T_\alpha \cap V \neq \emptyset$ , choose  $\psi(\alpha, V) \geq \alpha$  with  $x_{\psi(\alpha, V)} \in V$ , and define  $u_{(\alpha, V)} = x_{\psi(\alpha, V)}$ . The map  $\psi: D \rightarrow A$  is final, so  $u$  is a subnet of  $x$ .

For  $S \subseteq E$ , either  $S \in \mathcal{U}$  or  $E \setminus S \in \mathcal{U}$ . In the first case, after any index of the form  $(\alpha, S)$ , every later value of  $u$  lies in  $S$ ; the second case is identical with the complement. Hence  $u$  is universal.  $\square$

**Lemma 2** (Arbitrary common eventual refinement). *Let  $(v^j: B_j \rightarrow E)_{j \in J}$  be a nonempty family of nets with the same eventuality ultrafilter  $\mathcal{U}$ . Fix  $j_* \in J$ . There is a net  $w$  which is a subnet of  $v^{j_*}$  and an eventual subnet of  $v^j$  for every  $j \in J$ .*

*Proof.* For  $b \in B_{j_*}$ , write

$$R_j(b) = \{v_c^j : c \geq b\}.$$

Because  $v^j$  is eventually in its own tail range,  $R_j(b) \in \mathcal{U}$ .

Let  $D$  consist of pairs

$$d = (F, (b_j)_{j \in F}),$$

where  $F \subseteq J$  is finite,  $j_* \in F$ , and  $b_j \in B_j$ . Order  $D$  by

$$(F, (b_j)) \preceq (G, (c_j)) \iff F \subseteq G \text{ and } b_j \leq c_j \quad (j \in F).$$

The set  $D$  is directed: for two constraints take the union of their finite sets and, on every overlapping coordinate, an upper bound in  $B_j$ .

For every  $d \in D$ , the finite intersection

$$Q_d = \bigcap_{j \in F} R_j(b_j)$$

belongs to  $\mathcal{U}$  and is nonempty. Choose  $w_d \in Q_d$ . Fix  $j \in J$  and let  $D_j = \{d \in D : j \in F_d\}$ . This is a tail of  $D$ . For  $d \in D_j$ , choose  $\theta_j(d) \in B_j$  such that

$$\theta_j(d) \geq b_j(d) \quad \text{and} \quad v_{\theta_j(d)}^j = w_d.$$

The map  $\theta_j: D_j \rightarrow B_j$  is final. Indeed, for  $b_0 \in B_j$ , choose a constraint  $d_0$  containing  $j$  with  $b_j(d_0) = b_0$ . Whenever  $d \succeq d_0$ , one has  $\theta_j(d) \geq b_j(d) \geq b_0$ . Thus  $w|_{D_j}$  is a subnet of  $v^j$ .

Since  $j_*$  belongs to every  $F_d$ ,  $D_{j_*} = D$ ; hence  $w$  itself is a subnet of  $v^{j_*}$ . For every other  $j$ , the same conclusion holds on the tail  $D_j$ , so  $w$  is an eventual subnet of  $v^j$ .  $\square$

## 5 Arbitrary-family diagonal principle

**Theorem 3.** *Let  $E$  be a nonempty set, let  $x: A \rightarrow E$  be a net, and let  $(\mathcal{P}_i)_{i \in I}$  be a family of properties of nets indexed by an arbitrary set  $I$ . Assume:*

- (i) *each  $\mathcal{P}_i$  is preserved under passing to eventual subnets;*
- (ii) *for every  $i \in I$  and every subnet  $y$  of  $x$ , there is a further subnet of  $y$  satisfying  $\mathcal{P}_i$ .*

*Then  $x$  has a subnet satisfying  $\mathcal{P}_i$  for every  $i \in I$ .*

*Proof.* The conclusion is immediate if  $I = \emptyset$ , so assume  $I \neq \emptyset$ . By the universal-subnet lemma, choose a universal subnet  $u$  of  $x$ , and let  $\mathcal{U} = \mathfrak{F}_u$ .

For each  $i \in I$ , hypothesis (ii), applied to  $u$ , gives a subnet  $v^i$  of  $u$  satisfying  $\mathcal{P}_i$ . Every subnet of a universal net is universal: it is eventually in whichever of  $S$  and  $E \setminus S$  eventually contains  $u$ . Moreover,  $\mathfrak{F}_{v^i} \supseteq \mathfrak{F}_u = \mathcal{U}$ . Both sides are ultrafilters, so

$$\mathfrak{F}_{v^i} = \mathcal{U} \quad (i \in I).$$

Apply the common-refinement lemma to the family consisting of  $u$  and all  $v^i$ , taking  $u$  as the distinguished member. We obtain a net  $z$  which is a subnet of  $u$  and an eventual subnet of every  $v^i$ . Since  $v^i$  satisfies  $\mathcal{P}_i$  and  $\mathcal{P}_i$  is preserved under eventual subnets,  $z$  satisfies  $\mathcal{P}_i$  for every  $i \in I$ . Finally, the composition of final maps is final, so the subnet  $z$  of  $u$  is a subnet of  $x$ .  $\square$

## 6 Consequences

**Corollary 4** (Tychonoff). *Assuming Choice, an arbitrary product of compact spaces is compact.*

*Proof.* Let  $x$  be a net in  $\prod_{i \in I} X_i$ . For each  $i$ , let  $\mathcal{P}_i$  mean that the  $i$ th coordinate net converges. Compactness of  $X_i$  says that every subnet has a further subnet satisfying  $\mathcal{P}_i$ , and coordinate convergence is preserved under subnets. The theorem produces one subnet converging in every coordinate, hence in the product topology.  $\square$

**Corollary 5** (Banach–Alaoglu). *For every Banach space  $X$ , the dual unit ball is weak-star compact.*

*Proof.* Given a net in  $B_{X^*}$  and  $x \in X$ , let  $\mathcal{P}_x$  mean that the scalar evaluation net at  $x$  converges. Every subnet has a further subnet satisfying  $\mathcal{P}_x$ , because its values lie in a compact scalar disk. The theorem gives a subnet converging pointwise on all of  $X$ . Its pointwise limit is linear and bounded by one, hence belongs to  $B_{X^*}$ , and the subnet converges to it in the weak-star topology.  $\square$

## 7 Verification and limitations

The proof matches the final-map definition of subnet used in the source. In the common-refinement lemma, the distinguished component is refined on the entire index set; each other component is refined on a tail, exactly matching the heredity hypothesis. No countability assumption enters.

The result is nonconstructive. It uses the ultrafilter lemma and additional set-indexed selections, all available in ZFC. No assertion is made about the corresponding statement in ZF alone.

The run's lightweight indexes and a bounded web search on July 22, 2026 were checked using arXiv:2304.04189, the exact title and author, phrases from the source question, and close variants involving universal nets and common ultrafilters. No later explicit resolution was found. Since universal nets are classical, the exact theorem may nevertheless be folklore; novelty confidence is moderate pending specialist review.

## References

- [1] Y. Azouzi, *A diagonal principle for nets*, arXiv:2304.04189 (2023).
- [2] J. L. Kelley, *General Topology*, Van Nostrand, 1955.